

Builder Journal

Weekly reflection on learning, building, and growth

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Program: Bachelor of Science in Artificial Intelligence

Duration: 2023–2027

Week: 1

Q1

Why Did I Choose AI Engineering?

I chose AI Engineering because artificial intelligence is becoming an important part of today's world. Technology is growing very quickly, and I believe that we need to improve our skills to keep up with this change.

AI Engineering allows us to build useful systems that can solve real-life problems. AI can make our lives easier, reduce the time required to complete tasks, and help people become more productive. It is not only a field with a strong future scope, but it is also a field where we can create solutions that can positively affect people's daily lives.

Since I am already studying Artificial Intelligence, AI Engineering gives me the opportunity to turn my theoretical knowledge into practical applications. I want to learn how AI models, APIs, tools, and software can be combined to create complete and useful products.

Q2

What Type of AI Products Do I Want to Build?

I want to build AI products that can be used by ordinary people in their daily lives. I am interested in creating applications related to personal planning, productivity, education, and religion.

For example, I previously created a simple application called Tadhkir. In the future, I want to integrate AI into this application so that it can collect and provide authentic religious information. However, accuracy and authenticity would be very important for such a product, because religious information should always come from trusted sources.

I also want to build products that do not make people completely dependent on AI. Instead, I want AI to guide users, support their decisions, and help them become more organized and independent.

Q3

What Was My Biggest Learning This Week?

My biggest learning this week was understanding how AI applications are built and how APIs are integrated into software. Before this week, I had basic knowledge about AI models, but I did not fully understand how we could connect a model to an application.

During this week, I learned how APIs work and how they can be used to build simple AI-powered chatbots. I also learned about LLM applications, AI agents, tool calling, memory, planning, and multi-agent systems.

Another important lesson was that I can learn difficult things when I remain consistent and dedicated. This week gave me exposure to new concepts and helped me expand my knowledge. Writing this journal also gave me a chance to reflect on my progress — it helped me understand what I learned, where I struggled, and what I should improve in the future.

Q4

What Challenges Did I Face?

One of the main challenges I faced was designing the user interface of my AI workspace application. Although I have some knowledge of CSS, placing different elements in the correct positions was difficult for me. Sometimes the layout did not look balanced, and I was unsure about which colors, spacing, and design style I should choose.

Q5

How Did I Solve These Challenges?

To improve the user interface, I used ChatGPT to get design ideas and suggestions. It helped me understand how the application layout could look and which color palette would be suitable.

I did not directly copy everything. I used the suggestions as guidance and then made changes according to my application. I tested different layouts, adjusted the CSS, and improved the placement of elements step by step.

What Are My Goals for Week 2?

My main goal for Week 2 is to become more dedicated and consistent in my learning. I want to manage my time better and focus more deeply on the concepts instead of only completing tasks.

My technical goal is to learn how to build AI agents. I want to understand how an agent uses tools, memory, and planning to complete tasks. I also want to practice building a simple agent-based application.

In Week 2, I want to:

- Understand the basic architecture of AI agents
- Learn how tool calling works
- Explore memory and planning in agents
- Improve my Python and API integration skills
- Continue improving my GitHub and project documentation
- Build a simple but useful AI agent

"Every small step in learning is still a step forward." — End of Week 1 Entry